

535514: Reinforcement Learning

Lecture 13 — Inverse RL

Ping-Chun Hsieh

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Announcement

- Today's lecture: 10:10am-11:20am
- Poster presentations
 - 6/3 (Tuesday), schedule TBA
 - Template: <https://www.overleaf.com/read/djsjrnkzrjggq#ff79ae>
 - Reminder: Printing itself can take up to 1 day. Please manage to print your poster early

What We Have Learned So Far

- ▶ We have learned a bunch of **policy-based**, **value-based**, *and* **model-based** RL algorithms
- ▶ We assume that either (i) *we have “rewards” from the environment* or (ii) *we can design a proper reward function*
- ▶ However, what if both (i) and (ii) are NOT doable?

Could we learn / infer the reward function (and thereafter apply any RL method)?

Why Is Reward Design Challenging?



"Reward hacking"

“...We assumed the score the player earned would reflect the informal goal of finishing the race, so we included the game in an internal benchmark designed to measure the performance of reinforcement learning systems on racing games.

However, it turned out that the targets were laid out in such a way that the reinforcement learning agent could gain a high score without having to finish the course. This led to some unexpected behavior when we trained an RL agent to play the game.”

Imitation Learning: 2 Major Paradigms

- Suppose we are given *expert demonstrations*.
How to learn from them?

$$(s_0, a_0, s_1, a_1, \dots, s_T, a_T, s_{T+1})$$

1. Direct imitation learning

- Copy the **actions** of the expert
- No reasoning about the outcomes of actions

2. Human imitation learning

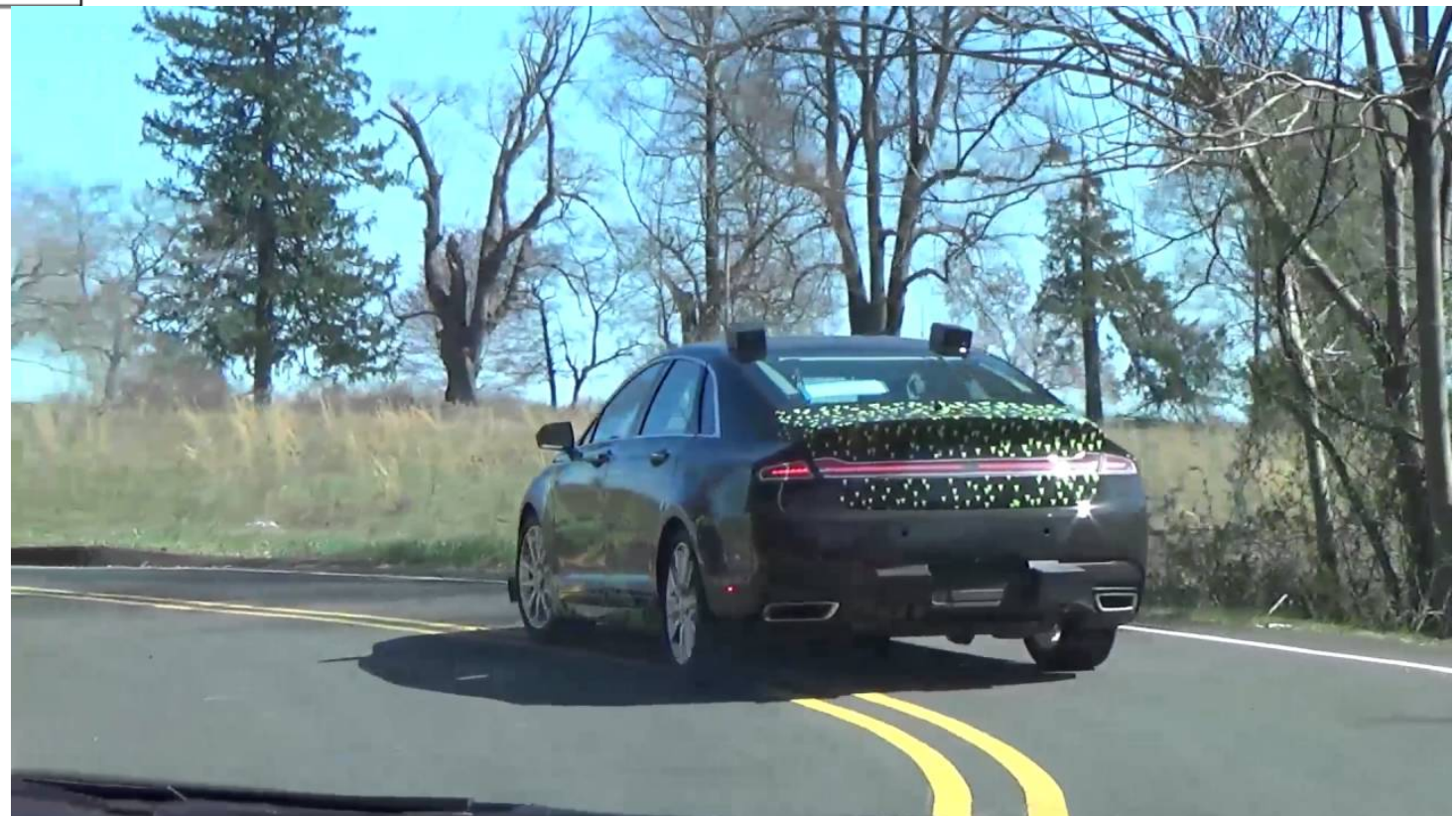
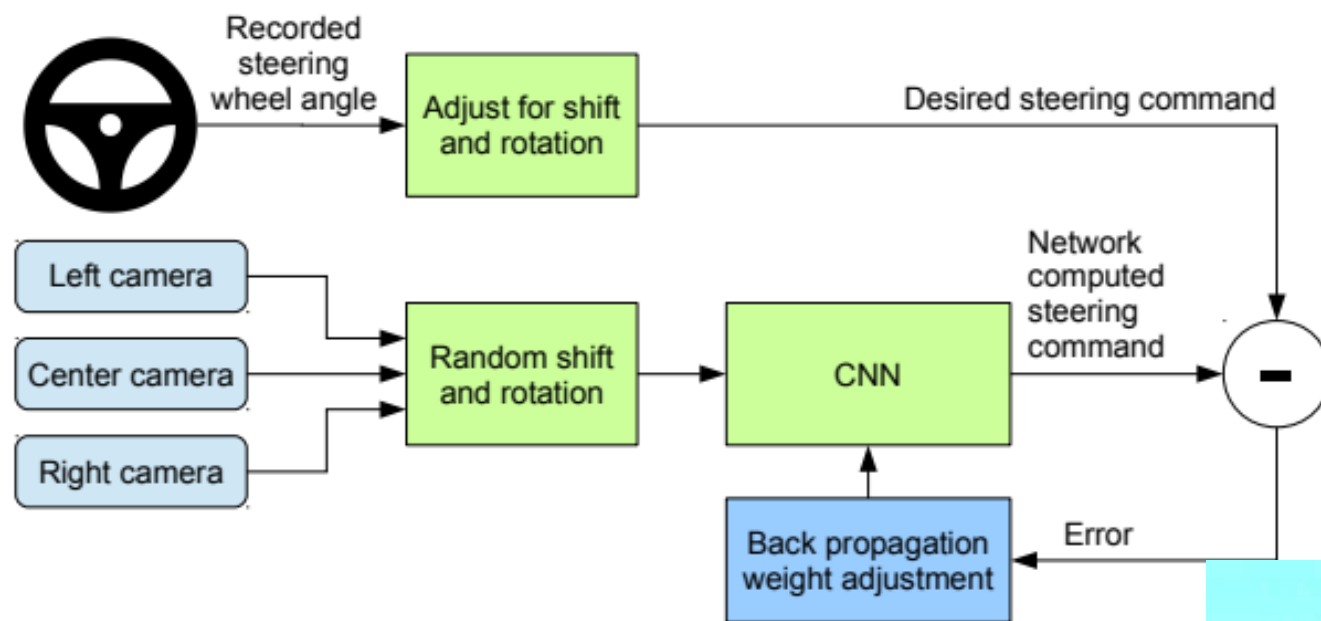
- Copy the **intent** of the expert
- May take very different actions from the expert

Inverse RL!

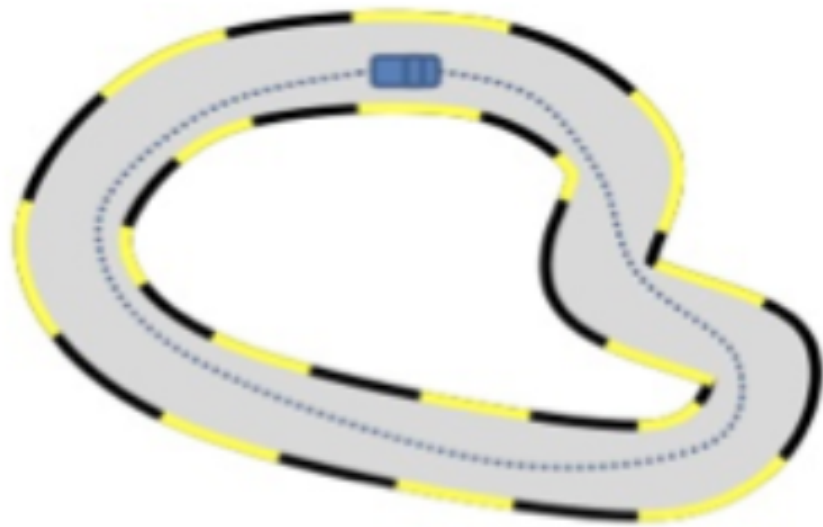


Direct Imitation Learning

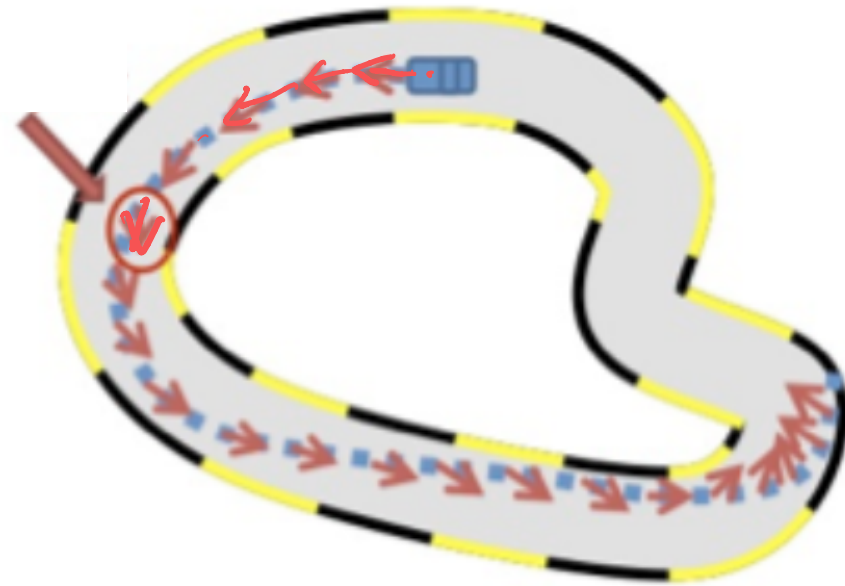
► Example: Self-driving cars



Direct Imitation Learning: Out-of-Distribution Issue



Expert Trajectories
(Training Distributions)

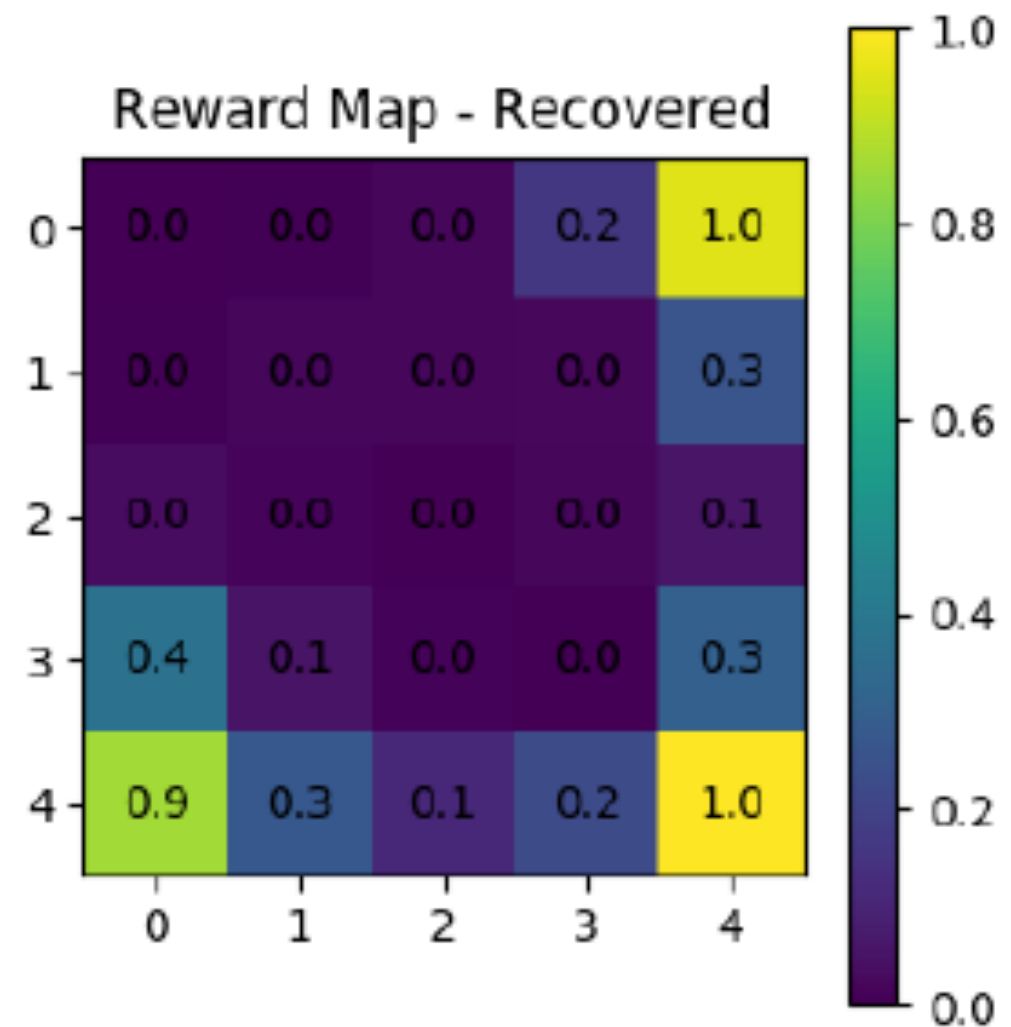
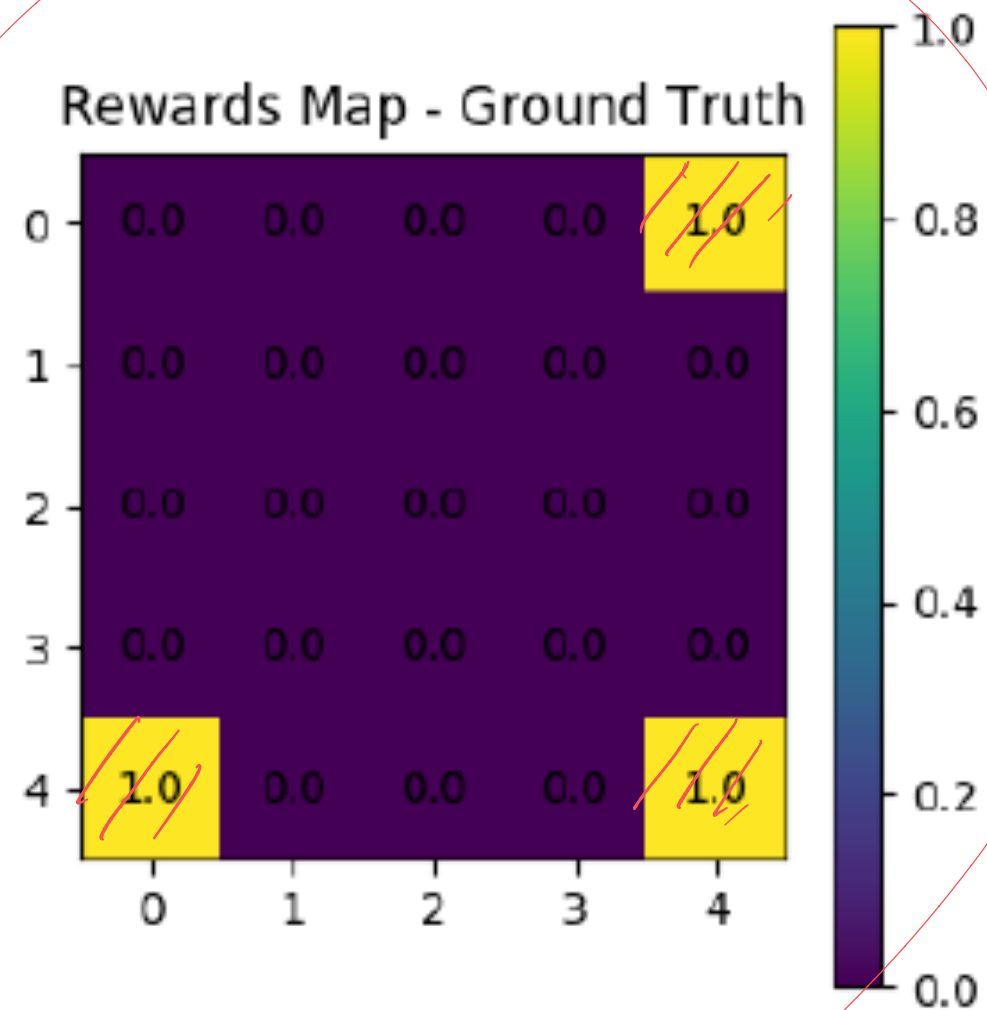


Direct Imitation Learning
(Testing Distributions)

- Makes mistakes, enter new states
- Cannot recover from new states

Inverse RL

- **Example:** Reward recovery in Gridworld

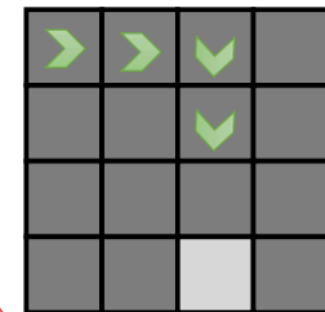
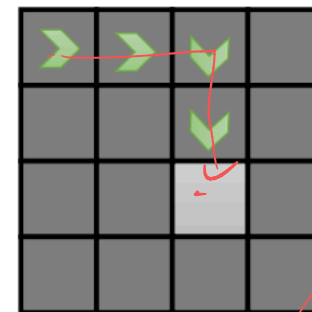
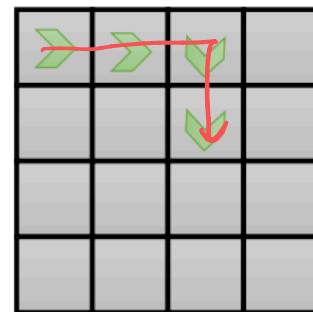
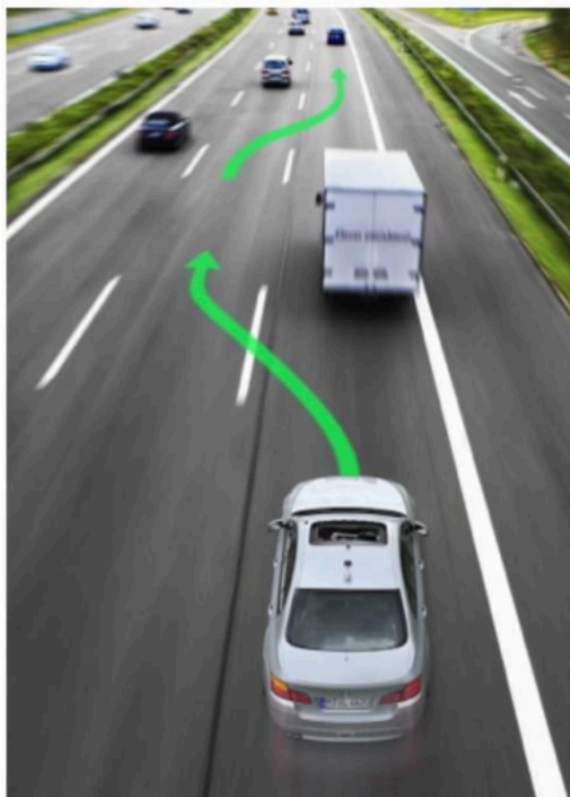


Inverse RL (Informal)

- ▶ Suppose the agent is in an MDP (S, A, P, γ)
- ▶ Suppose we are given expert demonstrations (under some unknown policy π_e)
- ▶ **Goal:** Infer the reward function R behind the expert actions solely from expert demonstrations (and thereafter learn a good policy)

First Attempt: Infer Rewards from Demonstrations

- **Example:** Human driving



Typically, “reward inference” is an underspecified problem

(Multiple reward functions can explain the same behavior)

What's reward $R(s, a)$?

Reward identifiability issue!

Rethinking Imitation?

$$V_{(\mu)}^{\pi_0} = E \left[\sum \gamma^t r_{t+1} \right]$$

$$= E \left[R(s, a) \right]$$

$s \sim d_{\mu}^{\pi_0}, a \sim \pi_0(\cdot | s)$

$$d_{\mu}^{\pi_0}(s, a)$$

Recall: Occupancy measure (or discounted state visitation)

$$d_{\mu}^{\pi}(s) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s | s_0, \pi) \right]$$

Occupancy measure

$$d_{\mu}^{\pi}(s, a) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s, a_t = a | s_0, \pi) \right]$$

(Q1) If $\pi_{\theta} = \pi_e$, then do we have $d_{\mu}^{\pi_{\theta}}(s, a) = d_{\mu}^{\pi_e}(s, a)$? Yes!

(Q2) If $d_{\mu}^{\pi_{\theta}}(s, a) = d_{\mu}^{\pi_e}(s, a)$, then do we have $V^{\pi_{\theta}}(\mu) = V^{\pi_e}(\mu)$? Yes!
(for "all" reward functions)

(Q3) If $d_{\mu}^{\pi_{\theta}}(s, a) = d_{\mu}^{\pi_e}(s, a)$, then do we have $\pi_{\theta} = \pi_e$? No!
But we have $\pi_{\theta}(\cdot | s) = \pi_e(\cdot | s)$ if $d_{\mu}^{\pi_e}(s) > 0$.

About (Q3): A Bijection Theorem

$$d^\pi(s, a) = 0.1$$

$$d^\pi(s, a') = 0.2$$

$$\pi'(a|s) = \frac{0.1}{0.1+0.2} = \frac{1}{3}$$

Theorem [Syed et al., 2008]: For any valid discounted state visitation distribution $d^\pi(s, a)$, define a policy $\pi'(a | s) := d^\pi(s, a) / \sum_{a' \in A} d^\pi(s, a')$.

Then, we have $d^{\pi'}(s, a) = d^\pi(s, a)$, for all (s, a) .

(In other words, the mapping from $d^\pi \rightarrow \pi$ is a bijection)

$$\pi'(a'|s) = \frac{0.2}{0.1+0.2} = \frac{2}{3}$$

- However, the above theorem does NOT implies that (Q3) holds

About (Q3): A Bijection Theorem

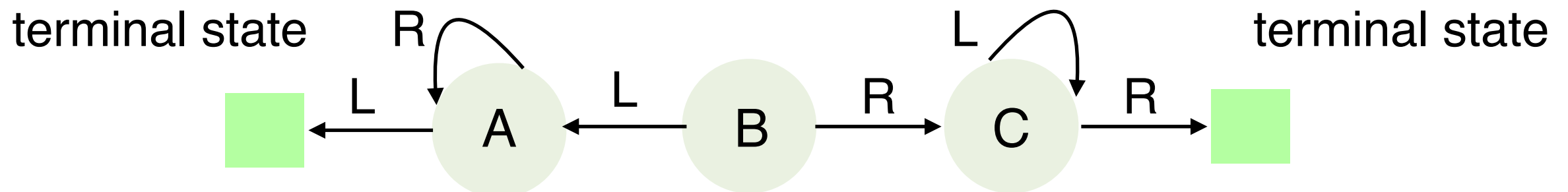
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Then, we have $d^{\pi'}(s, a) = d^\pi(s, a)$, for all (s, a) .

(In other words, the mapping from $d^\pi \rightarrow \pi$ is a bijection)

- Regarding (Q3), Bijection Theorem only implies that $\pi_\theta(\cdot | s) = \pi_e(\cdot | s)$ at those states with $d^{\pi_e}(s) > 0$

Example:



Inverse RL: Occupancy Measure Matching

Brian Ziebart et al., Maximum entropy inverse reinforcement learning, AAAI 2008

Jonathan Ho and S. Ermon, Generative adversarial imitation learning, NIPS 2016

Xiao et al., Wasserstein Adversarial Imitation Learning, arXiv 2019

Garg et al., IQ-Learn: Inverse soft-Q Learning for Imitation, NeurIPS 2021

Occupancy Measure Matching: Formulation

Recall: *Occupancy measure (or discounted state visitation)*

$$d_{\mu}^{\pi}(s) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s \mid s_0, \pi) \right]$$

$$d_{\mu}^{\pi}(s, a) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s, a_t = a \mid s_0, \pi) \right]$$

Claim: $V^{\pi}(\mu) = \sum_{(s,a)} d_{\mu}^{\pi}(s, a) R(s, a)$ (Why?)

Occupancy Measure Matching: Formulation

Recall: *Occupancy measure (or discounted state visitation)*

$$d_{\mu}^{\pi}(s) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s \mid s_0, \pi) \right]$$

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Claim: $V^{\pi}(\mu) = \sum_{(s,a)} d_{\mu}^{\pi}(s, a) R(s, a)$ (Why?)

Occupancy Measure Matching:

Find a policy π such that $d_{\mu}^{\pi}(s, a) = d^{\pi_e}(s, a), \quad \forall (s, a)$

Occupancy measure matching implies $V^{\pi}(\mu) = V^{\pi_e}(\mu)$

However, $d_{\mu}^{\pi}(s, a)$ can be hard to parameterize (why?)

Occupancy Measure Matching (OMM)

Direct Occupancy Measure Matching

$$\min_{\pi \in \Pi} L(\pi) := D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e})$$

($D(\cdot, \cdot)$ is some distance)

(d_{μ}^{π} could be hard to express!)

↕
Dual of each other!

$$\max_{R \in \mathcal{R}} \min_{\pi \in \Pi} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

OR

$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

(Easier for training!)

Apprenticeship Learning (APPLE)

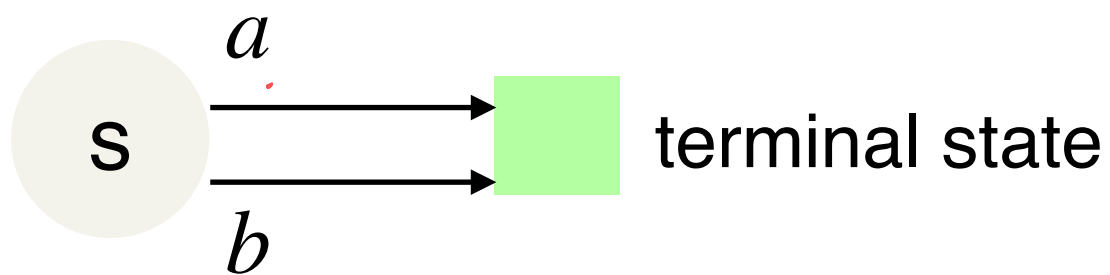
A Motivating Example: Connecting OMM & APPLE

$$\min_{\pi \in \Pi} L(\pi) := D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e})$$

$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:= L(\pi, R)} \right]$$

Handwritten notes above the equation: $\sum_{(s,a)} (d_{\mu}^{\pi_e}(s,a) - d_{\mu}^{\pi}(s,a)) \cdot R(s,a)$

Consider a simple 1-state, 2-action MDP



Suppose $\mathcal{R} = \mathbb{R}^2$

Handwritten notes: $R(s, a)$ and $R(s, b)$ in red.

$$\pi_e(a|s) = \pi_e(b|s) = 0.5$$

Let's write down $R \in \mathcal{R}$ that maximizes $L(\pi, R)$ under a fixed π

For (s, a) with $d_{\mu}^{\pi}(s, a) > d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = -\infty$

For (s, a) with $d_{\mu}^{\pi}(s, a) < d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = +\infty$

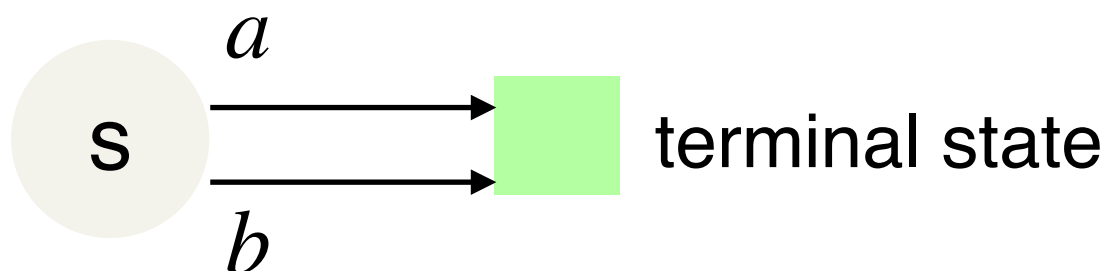
For (s, a) with $d_{\mu}^{\pi}(s, a) = d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = \text{any real number}$

A Motivating Example: Connecting OMM & APPLE

$$\min_{\pi \in \Pi} L(\pi) := D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e})$$

$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

Consider a simple 1-state, 2-action MDP



Suppose $\mathcal{R} = \mathbb{R}^2$

$$\pi_e(a|s) = \pi_e(b|s) = 0.5$$

Nice Property: Under $\mathcal{R} = \mathbb{R}^2$, the corresponding metric D is

$$D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e}) = \begin{cases} 0, & \text{if } d_{\mu}^{\pi}(s, a) = d_{\mu}^{\pi_e}(s, a), \forall (s, a) \\ \infty, & \text{otherwise} \end{cases}$$

0/1 distance

A Motivating Example: Connecting OMM & APPLE

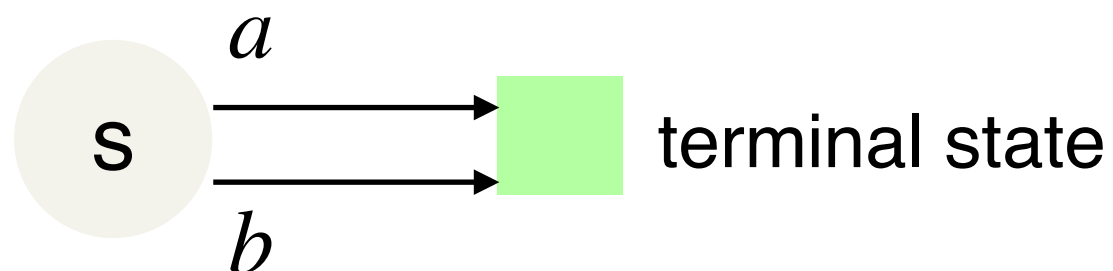
$$\min_{\pi \in \Pi} L(\pi) := D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e})$$

$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}}$$

$$\left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

$\sum (d_{\mu}^{\pi_e}(s, a) - d_{\mu}^{\pi}(s, a)) \cdot R(s, a)$

Consider a simple 1-state, 2-action MDP



Suppose $\mathcal{R} = \{ R \in \mathbb{R}^2 \mid \|R\|_{\infty} \leq 1 \}$

$$\pi_e(a|s) = \pi_e(b|s) = 0.5$$

Let's write down $R \in \mathcal{R}$ that maximizes $L(\pi, R)$ under a fixed π

For (s, a) with $d_{\mu}^{\pi}(s, a) > d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = -1$

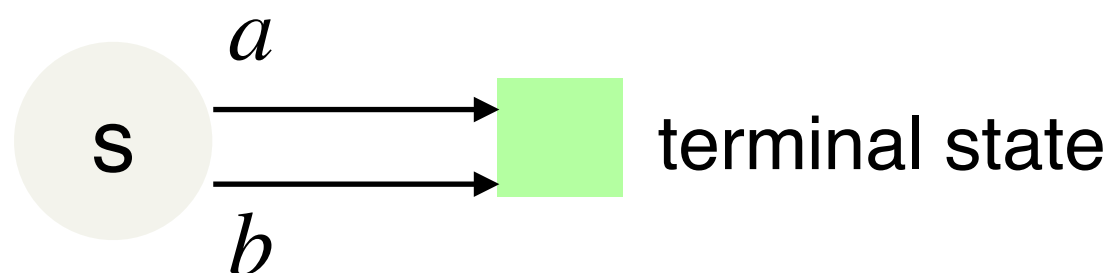
For (s, a) with $d_{\mu}^{\pi}(s, a) < d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = 1$

For (s, a) with $d_{\mu}^{\pi}(s, a) = d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = \text{any value between } -1 \text{ and } 1$

A Motivating Example: Connecting OMM & APPLE

$$\min_{\pi \in \Pi} L(\pi) := \underbrace{D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e})}_{\text{metric}} \longleftrightarrow \min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

Consider a simple 1-state, 2-action MDP



Suppose $\mathcal{R} = \left\{ R \in \mathbb{R}^2 \mid \|R\|_{\infty} \leq 1 \right\}$
 $\pi_e(a|s) = \pi_e(b|s) = 0.5$

Nice Property: Under $\mathcal{R} = \left\{ R \in \mathbb{R}^2 \mid \|R\|_{\infty} \leq 1 \right\}$, the metric D is

$$D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e}) = \sum_{(s,a)} \left| d_{\mu}^{\pi}(s, a) - d_{\mu}^{\pi_e}(s, a) \right|$$

(usually called “total variation distance”)

How to choose $\mathcal{R}$ to get some widely-used D ?

Example #1: Wasserstein Metric and APPLE

$$\min_{\pi \in \Pi} L(\pi) := W(d_{\mu}^{\pi}, d_{\mu}^{\pi_e})$$

(Wasserstein)



$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

$$\text{where } \mathcal{R} = \left\{ R \in \mathbb{R}^{\mathcal{S} \times \mathcal{A}} \mid \text{Lip}(R) \leq 1 \right\}$$

This is also known as the *Kantorovich-Rubenstein duality*

$$\left| R(s, a) - R(s', a') \right| \leq L \cdot \|(s, a) - (s', a')\|_2$$

Proof: https://courses.cs.washington.edu/courses/cse599i/20au/resources/L12_duality.pdf

Wasserstein Metric

Metric for random vectors

- ▶ $U : \Omega \rightarrow \mathbb{R}^d$: a random vector from the sample space Ω to $\mathbb{R}^d$
- ▶ For $1 \leq p < \infty$: $\|U\|_p := \left(\mathbb{E} [\|U(\omega)\|_p^p] \right)^{\frac{1}{p}}$

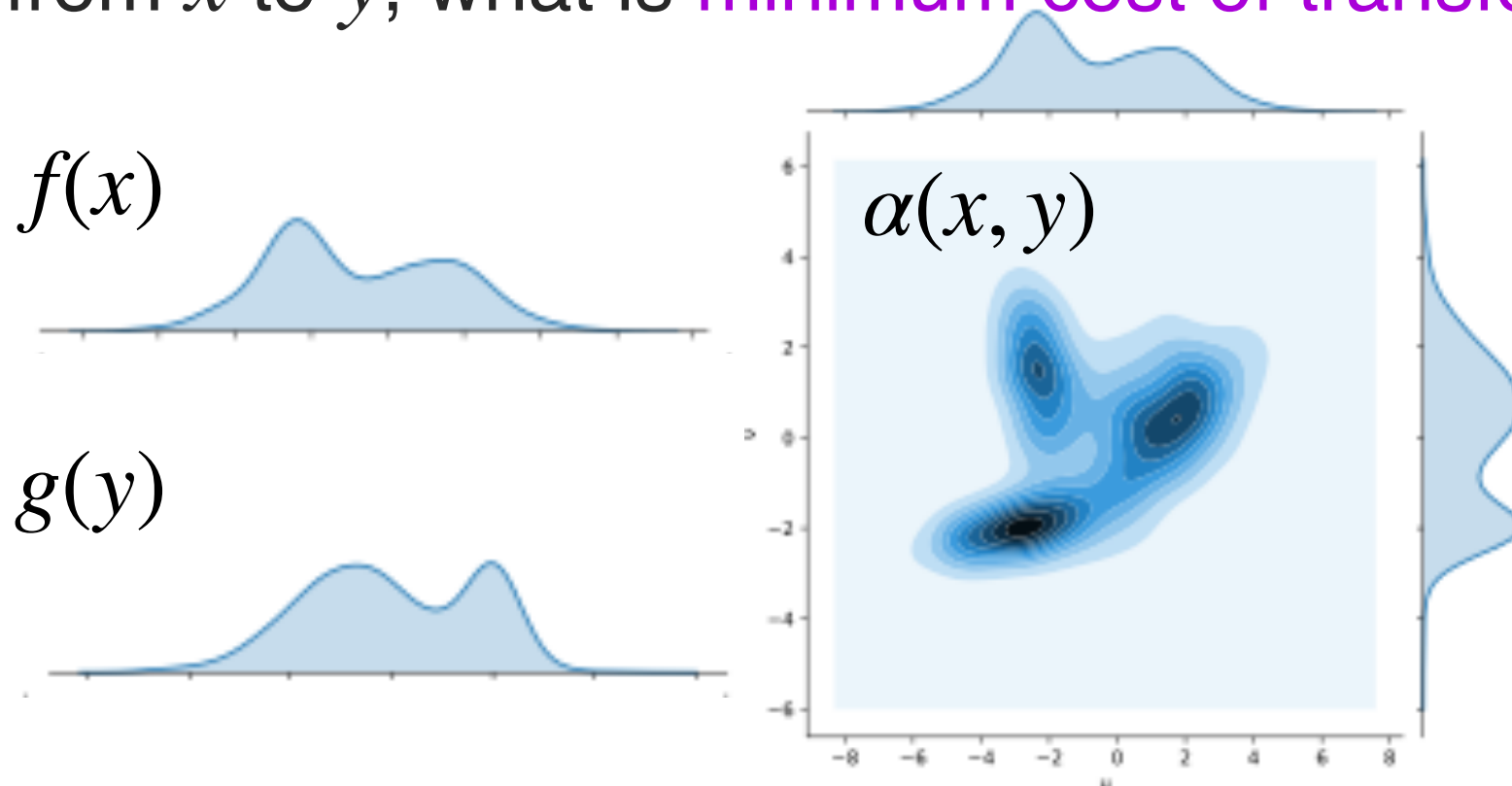
- ▶ **Wasserstein Metric**: For two CDFs F, G over the reals, the Wasserstein metric is defined as

$$d_p(F, G) := \inf_{(U, V): U \sim F, V \sim G} \|U - V\|_p$$

- ▶ Infimum is taken over all joint distributions of random variables (U, V) , whose marginal distributions are F, G

Intuition Behind Wasserstein Metric

- ▶ Also known as: optimal transport problem or earth mover's distance
- ▶ Given two density $f(x)$, $g(x)$ and a cost function $c(x, y)$ of moving mass from x to y , what is **minimum cost of transforming from $f(x)$ to $g(y)$** ?



Minimum cost

$$C^* := \inf_{\alpha} \int c(x, y) \alpha(x, y) dx dy$$

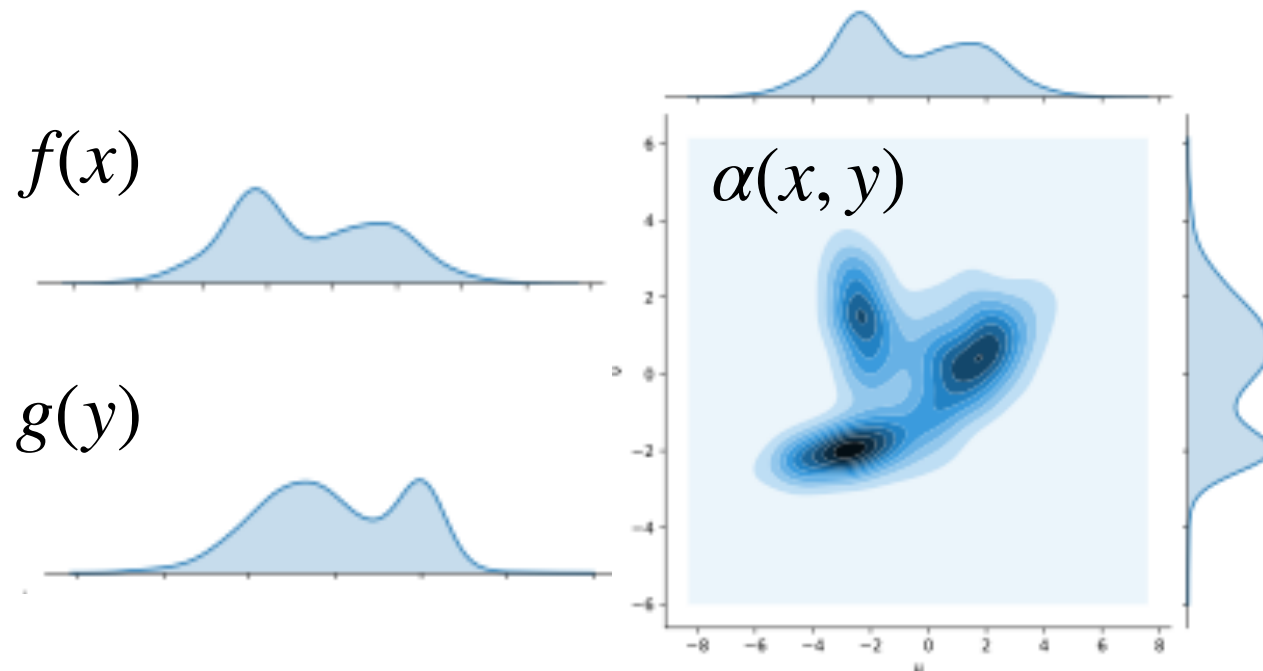
$\alpha(x, y)$: amount of mass to move from x to y
 $\alpha(x, y)$ describes a feasible transport plan if

$$\int \alpha(x, y) dy = f(x), \quad \int \alpha(x, y) dx = g(y)$$

Summary: Optimal Transport & Wasserstein Metric

Wasserstein $d_p(F, G) := \inf_{(U, V): U \sim F, V \sim G} \|U - V\|_p$

**Optimal
Transport
(OT)**



$c(x, y)$ = cost function of moving one unit of mass from x to y

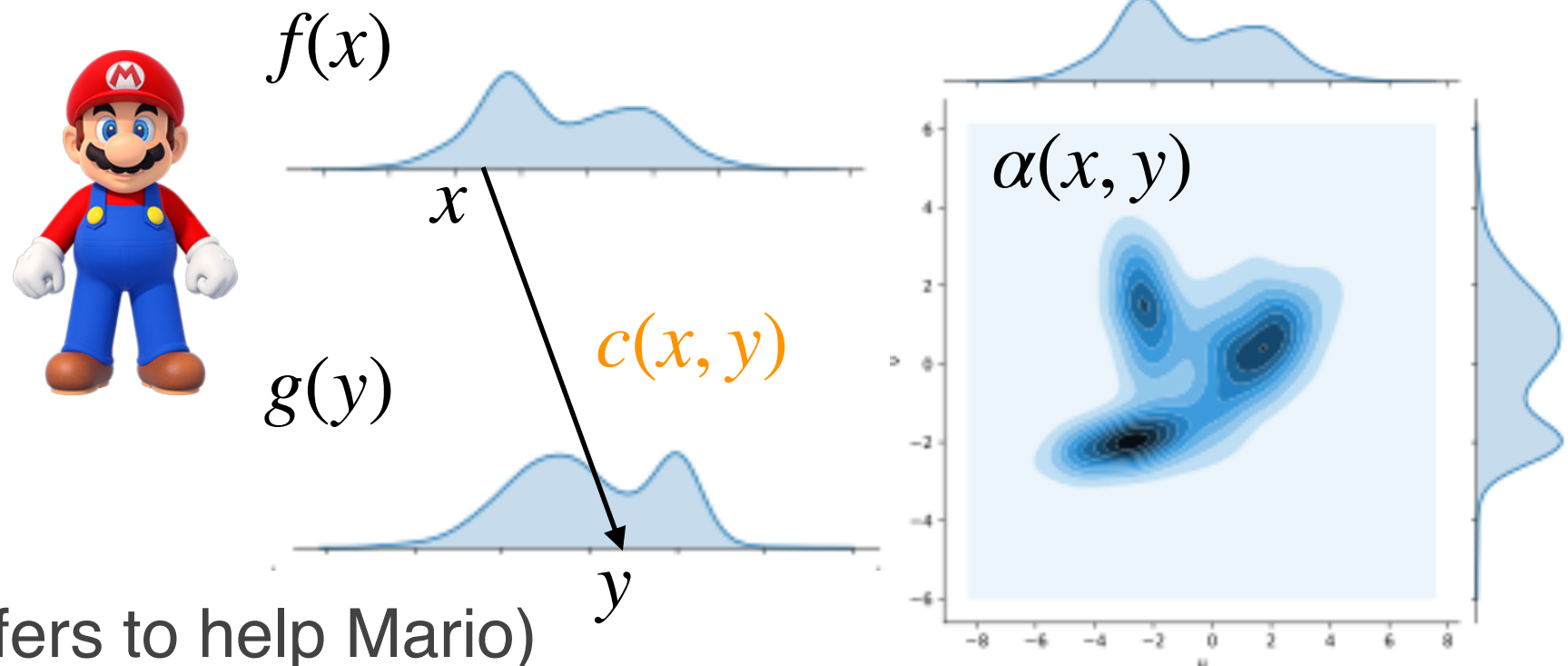
- OT can be written as an optimization problem:

$$\begin{aligned} & \min_{\alpha} \sum_{x, y} c(x, y) \alpha(x, y) \\ \text{subject to} \quad & (1) \sum_y \alpha(x, y) = f(x), \forall x \quad (2) \sum_x \alpha(x, y) = g(y), \forall y \\ & (3) \alpha(x, y) \geq 0, \forall x, y \end{aligned}$$

Duality of Optimal Transport: Economic Interpretation

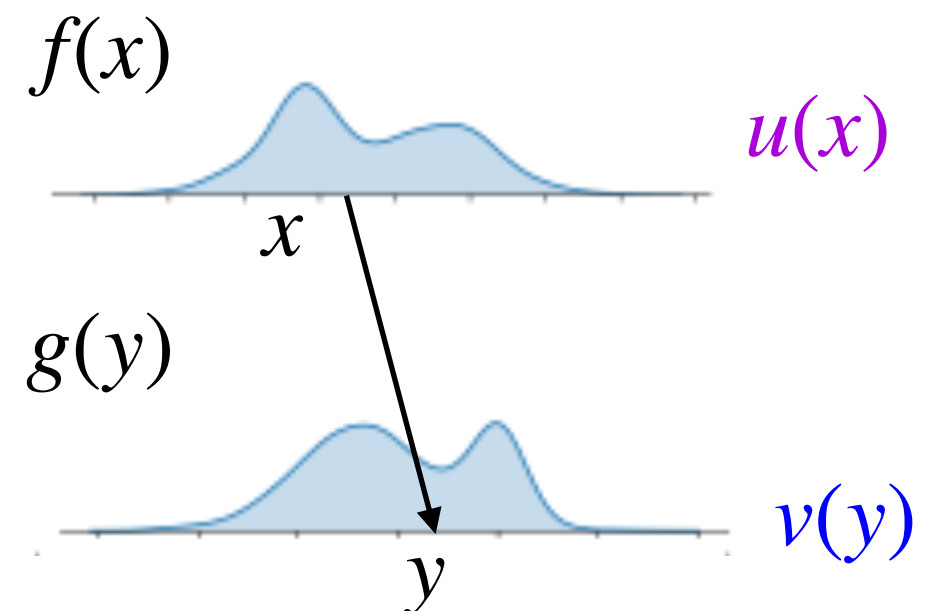
Primal Form of OT (Mario moving the earth by himself)

$c(x, y)$ = Mario's cost function
(for moving one unit of mass
from x to y)



Dual Form of OT (Luigi offers to help Mario)

Mario needs to pay Luigi $u(x)$
and $v(y)$ (for moving one unit
of mass from x to y)



Question: Under what condition would Mario ask for Luigi's help?

Duality of Optimal Transport (Formally)

► Primal Form of Optimal Transport

$$\begin{aligned} & \min_{\alpha} \sum_{x,y} c(x,y) \alpha(x,y) \\ \text{subject to } & (1) \sum_y \alpha(x,y) = f(x), \forall x \quad (2) \sum_x \alpha(x,y) = g(y), \forall y \\ & (3) \alpha(x,y) \geq 0, \forall x, y \end{aligned}$$

► Dual Form of Optimal Transport

$$\begin{aligned} & \max_{u,v} \mathbb{E}_{x \sim f(x)}[u(x)] + \mathbb{E}_{y \sim g(y)}[v(y)] \\ \text{subject to } & u(x) + v(y) \leq c(x,y), \forall x, y \end{aligned}$$

The dual form looks exactly like APPLE!

- Both forms lead to the same optimal values (called “strong duality”)

Example #2: Generative Adversarial Imitation Learning (GAIL)

- **Recall**: Dual Form of Optimal Transport

$$\max_{u,v} \mathbb{E}_{x \sim f(x)}[u(x)] + \mathbb{E}_{y \sim g(y)}[v(y)]$$

subject to $u(x) + v(y) \leq c(x, y), \forall x, y$

$D_\phi(s, a)$: A **binary classifier** that predicts the probability of the event that “the observed (s, a) is drawn from π ”

Let's choose the following:

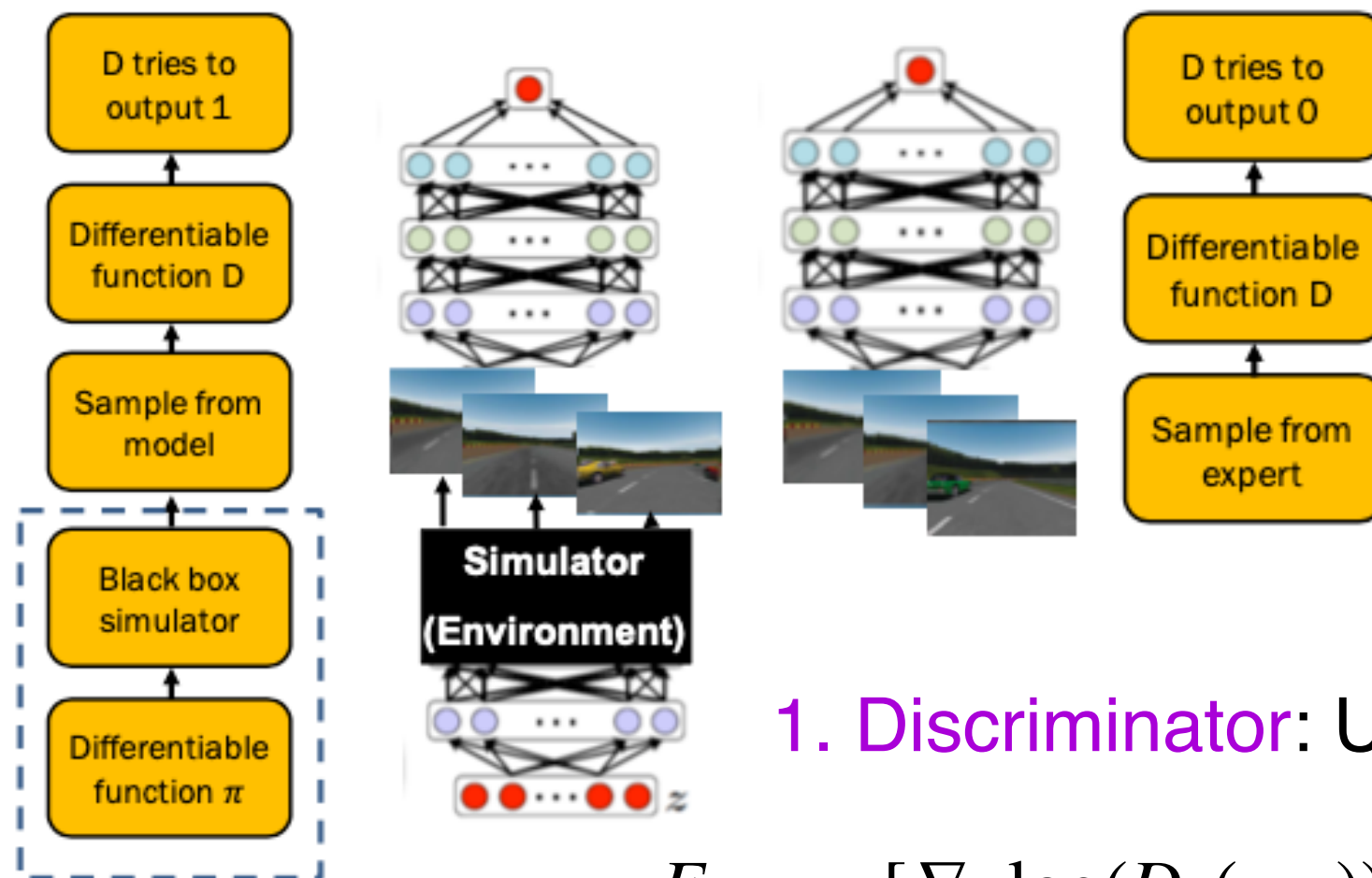
(1) $f(x) \equiv d_\mu^\pi(s, a)$

(2) $g(y) \equiv d_\mu^{\pi_e}(s, a)$

(3) $u(x) \equiv \log(D_\phi(s, a))$

(4) $v(y) \equiv \log(1 - D_\phi(s, a))$

GAIL: Discriminator and Generator



1. Discriminator: Update ϕ by

$$E_{(s,a) \sim d_{\mu}^{\pi}}[\nabla_{\phi} \log(D_{\phi}(s, a))] + E_{(s,a) \sim d_{\mu}^{\pi_e}}[\nabla_{\phi} \log(1 - D_{\phi}(s, a))]$$

2. Generator: Use any RL algorithm with reward function $\log(D_{\phi}(s, a))$

A Comparison Between Wasserstein AIL and GAIL

Expert state-action pairs (projected onto a 2D board via PCA)

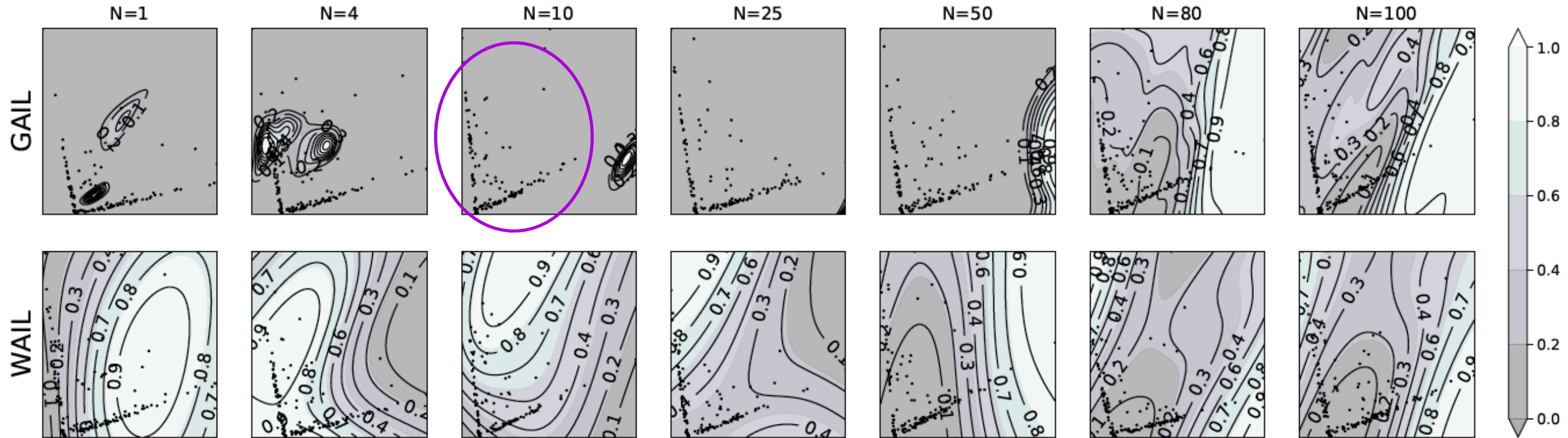


Figure 2: Reward surfaces of WAIL and GAIL on *Humanoid* with respect to different expert data sizes.

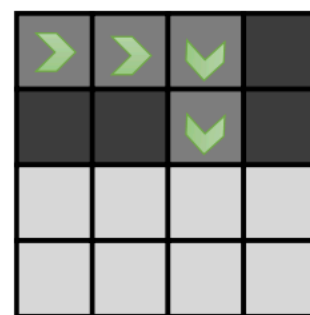
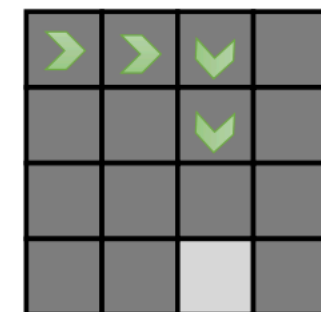
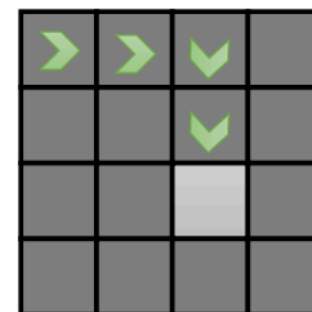
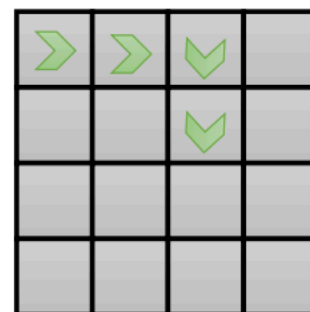
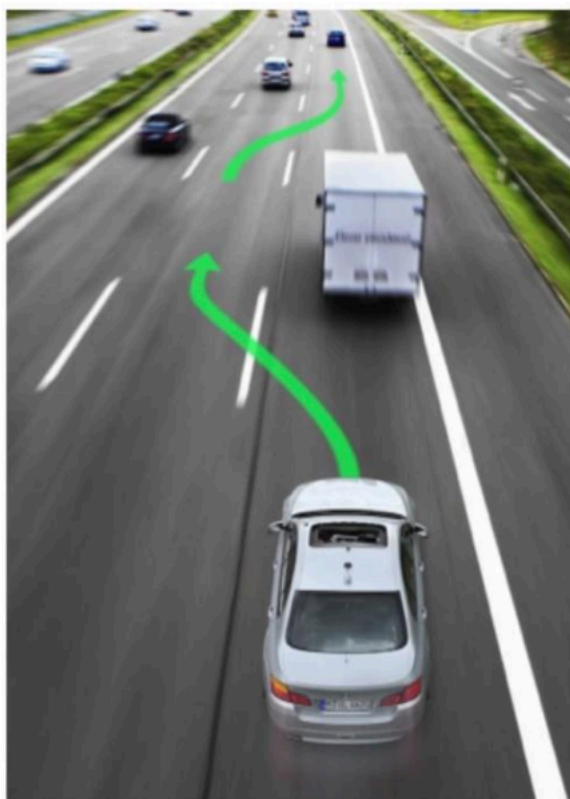
Occupancy Measure Matching via Apprenticeship Learning

$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{(s,a) \sim d_{\mu}^{\pi_e}}[R(s, a)] - E_{(s,a) \sim d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

Key Idea: By choosing different “reward function classes $\mathcal{R}$ ”, we obtain various OMM approaches!

Recap: Infer Rewards from Demonstrations

- **Example:** Human driving



Typically, “reward inference” is an underspecified problem

(Multiple reward functions can explain the same behavior)

What's reward $R(s, a)$?

Reward identifiability issue!

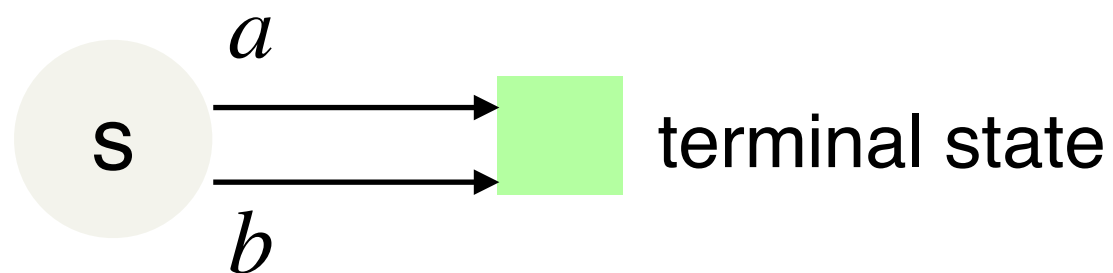
Have we solved the identifiability issue by OMM or APPLE?

Actually not!

A Motivating Example: Connecting OMM & APPLE

$$\min_{\pi \in \Pi} L(\pi) := D(d_{\mu}^{\pi}, d_{\mu}^{\pi_e}) \longleftrightarrow \min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{d_{\mu}^{\pi_e}}[R(s, a)] - E_{d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} \right]$$

Consider a simple 1-state, 2-action MDP



Suppose $\mathcal{R} = \mathbb{R}^2$

$$\pi_e(a|s) = \pi_e(b|s) = 0.5$$

Let's write down $R \in \mathcal{R}$ that maximizes $L(\pi, R)$ under a fixed π

For (s, a) with $d_{\mu}^{\pi}(s, a) > d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = +\infty$

For (s, a) with $d_{\mu}^{\pi}(s, a) < d_{\mu}^{\pi_e}(s, a)$: $R(s, a) = -\infty$

For (s, a) with $d_{\mu}^{\pi}(s, a) = d_{\mu}^{\pi_e}(s, a)$: $R(s, a)$ can be arbitrary

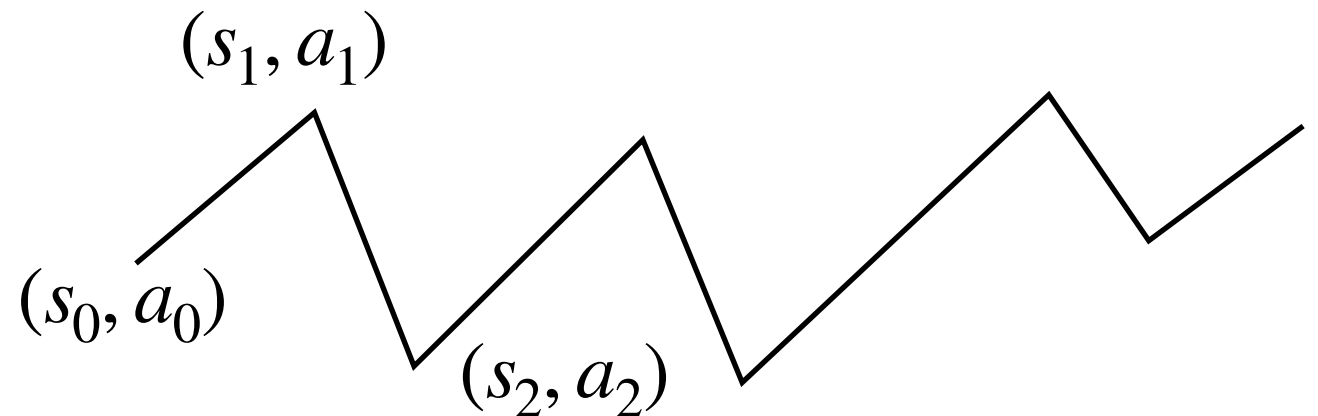
Reward function is
not unique!!

Max-Entropy Principle for IRL

- ▶ A policy π induces distribution over trajectories (denoted by $p(\tau)$)

$$\tau = (s_0, a_0, s_1, a_1 \dots)$$

$$d(s, a; \tau) := \sum_t \gamma^t \mathbb{I}(s_t = s, a_t = a)$$



“Occupancy measure matching”: Find a policy π that satisfies

$$d_{\mu}^{\pi}(s, a) \equiv \sum_{\tau} p(\tau) d(s, a; \tau) = d_{\mu}^{\pi_e}(s, a)$$

- ▶ **Maximum Entropy Principle**: The probability distribution that “best” represents the current state of knowledge is the one with the largest entropy

Max-Entropy IRL

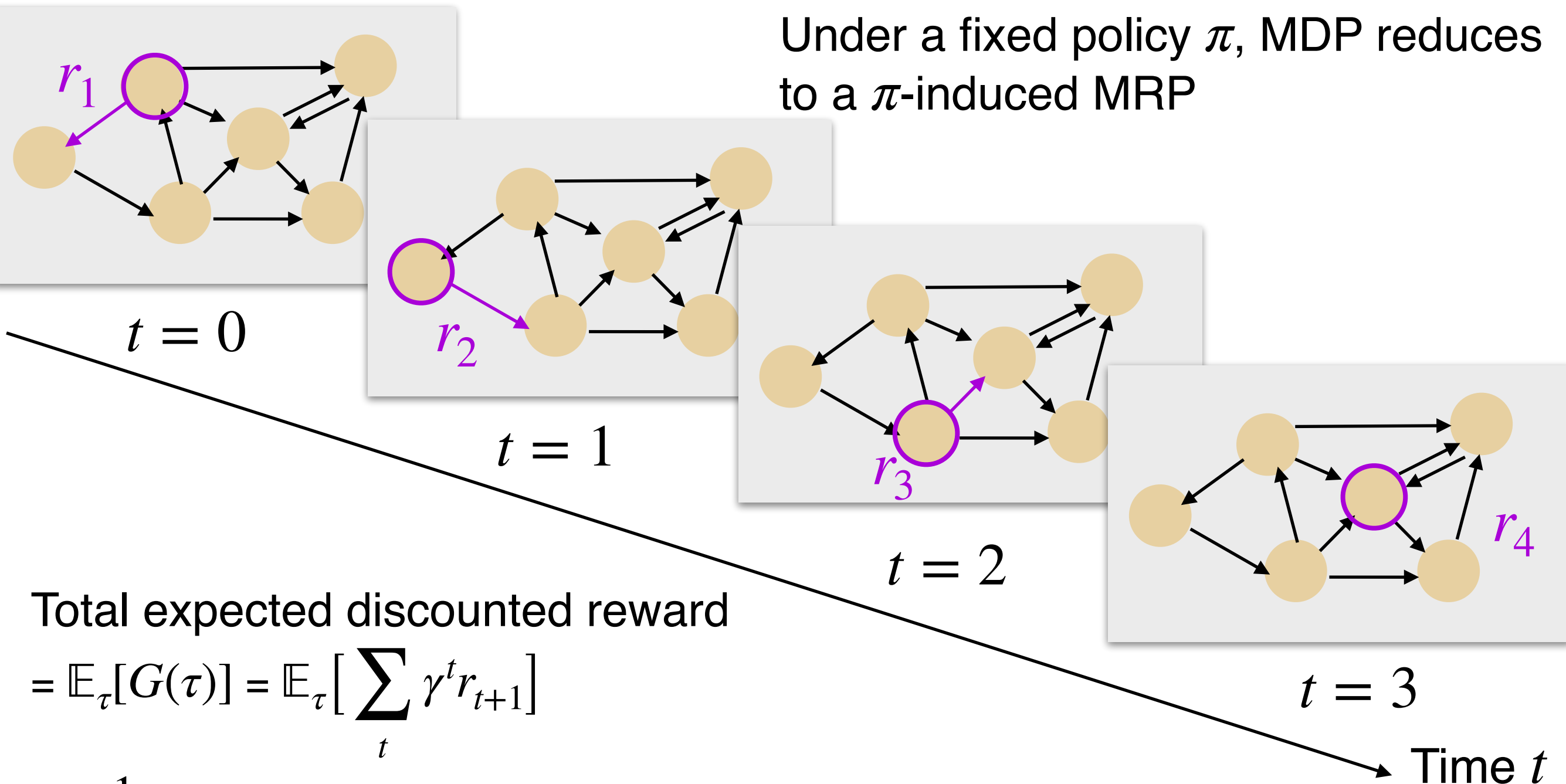
- ▶ Applying Maximum Entropy principle to IRL:

$$\begin{aligned} \max_{p(\tau)} \quad & - \sum_{\tau} p(\tau) \log p(\tau) \equiv H(p(\tau)) \\ \text{subject to} \quad & \sum_{\tau} p(\tau) d(s, a; \tau) = d_{\mu}^{\pi_e}(s, a), \forall (s, a) \\ & \sum_{\tau} p(\tau) = 1 \end{aligned}$$

- ▶ **Intuition:** Choose the most “uncertain” distribution $p(\tau)$ among those matching the occupancy measure

Review: Time vs Distribution Domain

Under a fixed policy π , MDP reduces to a π -induced MRP



Total expected discounted reward

$$= \mathbb{E}_{\tau}[G(\tau)] = \mathbb{E}_{\tau}\left[\sum_t \gamma^t r_{t+1}\right]$$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}, a \sim \pi_{\theta}}[R(s, a)]$$

Occupancy Measure Matching via Apprenticeship Learning (With Regularization)

$$\min_{\pi \in \Pi} \max_{R \in \mathcal{R}} \left[\underbrace{\left(E_{(s,a) \sim d_{\mu}^{\pi_e}}[R(s, a)] - E_{(s,a) \sim d_{\mu}^{\pi}}[R(s, a)] \right)}_{:=L(\pi, R)} - H(\pi) + \psi(R) \right]$$

where $H(\pi) := E \left[\sum_t -\gamma^t \log \pi_t(a_t | s_t) \right]$ is the discounted causal entropy

$\psi(R)$ is a regularizer for the reward function

Key Idea: Adding both entropy and reward regularization can mitigate the identifiability issue (basically tie breaking)